

YIXUAN LI

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Education

Southern University of Science and Technology (SUSTech)

Aug. 2022 – Jun. 2026 (Expected)

Bachelor of Science in Data Science

Shenzhen, China

- **GPA:** 3.79/4.00 (3.92/4.00 in Junior Year)
- **Mathematics Courses:** Calculus, Linear Algebra, Mathematical Analysis, Discrete Mathematics and Its Applications, Probability Theory, Mathematical Statistics, Statistical Linear Models, Operational Research and Optimization
- **CS Courses:** Introduction to Computer Programming, Data Structures and Algorithm Analysis, Principles of Database Systems, Distributed Storage and Parallel Computing, Advanced Natural Language Processing, Artificial Intelligence

University of California, San Diego (UCSD)

Mar. 2025 – Jun. 2025

University and Professional Studies (UPS) Program

San Diego, CA, United States

- **GPA:** 4.00/4.00
- **Courses:** Machine Learning Algorithms, AI Search and Reasoning, Topics of Data Science: Trustworthy Machine Learning

Research Interests

- **Trustworthy Machine Learning:** Uncertainty Estimation & Calibration, Hallucination Detection, Model Interpretability, LLM Alignment, Generative AI
- **AI for Science (Biomedical):** Multimodal Learning, Graph Neural Networks (GNNs), Computational Biology

Research Experience

SUSTech Machine Learning Lab | Advisor: [Prof. Hongxin Wei](#) | SUSTech

Shenzhen, China

Project: Multimodal Learning for Spatial Transcriptomics-to-Proteomics Prediction

Main Developer

Oct. 2025 – Present

- Designed a unified pipeline for the [STP Open Challenge](#) that predicts spatial protein expression from RNA in glioma tissue, integrating spatial graphs, molecular signals, and morphological cues in a biologically grounded framework.
- Built a [Multimodal Graph Attention Network](#) that fuses RNA embeddings with morphology features extracted from H&E patches using pathology foundation models, employing gated fusion to align molecular and tissue-level representations.
- Proposed biology-informed regularization strategies to enhance robustness and cross-slice generalization, including: (i) [cell-type abundance features](#) as an auxiliary modality, coupled with structure-aware similarity regularization to enforce local biological consistency, and (ii) [STRING-derived protein co-expression priors](#) incorporated as loss-level constraints.
- Achieved **Top-1 (Develop)** and **Top-2 (Test)** performance on the [public leaderboard](#), demonstrating robust cross-slice generalization (consistent Spearman correlation up to 0.56 across tissue slices), with fully reproducible [code](#) available.

Project: Benchmarking Uncertainty Quantification Methods in Large Language Models

Independent Researcher

Jun. 2025 – Oct. 2025

- Reproduced and extended a [semantic uncertainty evaluation pipeline](#) for LLMs, integrating a new method and adapting the codebase to handle model-format inconsistencies, enabling systematic comparisons across QA datasets (AUROC, AUARC, PRR).
- Conducted an extensive literature review on [uncertainty quantification in LLMs](#), connecting with [hallucination detection](#) and [calibration](#), and summarized insights in internal slide decks and short blog-style write-ups.

Jianqing Shi's Lab | Advisor: [Prof. Jianqing Shi](#) | SUSTech

Shenzhen, China

Project: Statistical Analysis of Mental Health Impact among Menopausal Women

Research Assistant

Sep. 2025 – Nov. 2025

- Analyzed large-scale survey data on menopausal women in Malaysia to examine the relationship between menopausal stages and mental health outcomes.
- Applied statistical methods including [Kruskal-Wallis test](#), [Fisher's exact test](#), [Chi-square test](#), [ANCOVA](#), [Cochran-Mantel-Haenszel test](#), [Paired t-tests](#), and [multiple imputation](#) for hypothesis testing and subgroup analysis.
- Identified stage-adjusted psychological risk factors and enhanced inference robustness in women's health research.
- Results reported internally due to data confidentiality.

Internship Experience

Machine Learning for Clinical Decision Support in Orthodontics

Jun. 2024 – Aug. 2024

Clinical Data Science Intern — Mentor: Dr. Ying Shan

Peking University Shenzhen Hospital

- Performed data preprocessing on clinical craniofacial imaging datasets, dental records, and treatment histories, which included cleaning, de-duplication, and standardization.
- Developed *linear and neural network models* based on craniofacial measurement variables to predict patient-specific tooth extraction plans. Achieved strong predictive performance, contributing to internal clinical adoption.

Personal Projects & Competitions

Concept Bottleneck in Generative Models: Reproduction Study | Core Member

Apr. 2025 – Jun. 2025

- Reproduced the *CVPR paper* on Post-hoc Concept Bottleneck frameworks: *Concept Bottleneck Autoencoder (CB-AE)*, *Concept Controller (CC)* using *StyleGAN*, replicating key quantitative and qualitative results from the original study.
- Conducted additional experiments on *Concept Extra-Interpolation*, *Entanglement*, and *Leakage* to evaluate model interpretability, robustness, and reliability under concept-level perturbations.
- Delivered a comprehensive reproduction package including *GitHub code*, *presentation slides*, and a public *technical blog*, earning **43.4/40** on the final project (**Top 1** in a UCSD class enrolling both undergraduate and graduate students).

Kaggle Competition: Fine-Tuning Gemma-2B for Chinese QA Dataset | Team Leader

Oct. 2024 – Jan. 2025

- Led a team to fine-tune *Gemma-2B* on a Chinese GPT-4 QA dataset, developing a complete pipeline from data preprocessing to LoRA-based fine-tuning in the *Kaggle competition*.
- Integrated *Retrieval-Augmented Generation (RAG)* for data augmentation and *LoRA-based PEFT* for parameter-efficient fine-tuning, resulting in improved dataset diversity, contextual grounding, and instruction-following performance.
- Evaluated model performance using *BLEU*, *ROUGE*, *METEOR*, and *BERTScore* metrics, and performed *cross-lingual robustness analysis* showing stable results across four languages. Published notebook can be viewed *here*.

Reinforcement Learning on FrozenLake Game Setting | Core Member

Oct. 2024 – Jan. 2025

- Implemented and evaluated the *Double DQN* and *Dueling DQN* algorithms on the FrozenLake environment, performing hyperparameter tuning and robustness analysis to achieve up to **98%** success rate in optimal path learning.
- Compiled a comprehensive report and structured a well-documented *GitHub code repository* to ensure reproducibility.

Awards

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| First Class of the Merit Student Scholarship, SUSTech (Top 5%, two-time recipient) | 2023 – 2025 |
| Third Class of the Merit Student Scholarship, SUSTech (Top 15%) | 2022 – 2023 |
| Outstanding Student Award, SUSTech (three-time recipient) | 2022 – 2025 |
| Second Prize China Undergraduate Mathematical Contest in Modeling (CUMCM), Guangdong Division | Oct. 2024 |
| Honorable Mention Mathematical Contest In Modeling (MCM) | May 2024 |

Leadership & Activities

Community Involvement

Oct. 2023 – Present

Volunteer

SUSTech

- Taught online academic enrichment classes for students in rural areas, supporting long-term educational outreach efforts.
- Contributed over **179 hours** of volunteer service.
- Awarded **Outstanding Volunteer** honors at both the university and School of Science levels.

Shuren College Media Center

Sep. 2023 – Aug. 2024

President

SUSTech

- Led a 15-member team producing digital communications and organizing college events across official media channels.
- Received the **Excellent Student Leader Award** and the **Outstanding College Organization Award**.

Skills

- Programming:** Python, R, SQL, Java, JavaScript, HTML
- Development Tools:** PyTorch, Git, Jupyter Notebook, LaTeX, Markdown, Hadoop
- Languages:** English (TOEFL 109: R28/L27/S27/W27), Mandarin (Native), Cantonese (Fluent)